

Name: _____

Double Award Science – Physics Unit 2 Foundation Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /60	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	G	F	E	D	C	Max UMS
2017	7	10	13	17	21	25
2018	5	9	13	17	22	27
2019	10	14	18	22	26	30
2022	6	11	16	21	27	33
2023	13	17	21	26	31	36

Surname	Centre Number	Candidate Number
Other Names		0



GCSE – **NEW**

3430U60-1



SCIENCE (Double Award)

Unit 6 – PHYSICS 2
FOUNDATION TIER

WEDNESDAY, 23 MAY 2018 – AFTERNOON

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	6	
2.	8	
3.	8	
4.	8	
5.	15	
6.	15	
Total	60	

ADDITIONAL MATERIALS

In addition to this examination paper, you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the continuation page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question **5(a)(ii)**.



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Equations

speed = $\frac{\text{distance}}{\text{time}}$	
acceleration [or deceleration] = $\frac{\text{change in velocity}}{\text{time}}$	$a = \frac{\Delta v}{t}$
acceleration = gradient of a velocity-time graph	
resultant force = mass $\times$ acceleration	$F = ma$
weight = mass $\times$ gravitational field strength	$W = mg$
work = force $\times$ distance	$W = Fd$
force = spring constant $\times$ extension	$F = kx$

SI multipliers

Prefix	Multiplier
m	1×10^{-3}
k	1×10^3
M	1×10^6



Answer all questions.

1. Road traffic accidents occur when a vehicle is unable to stop safely. The overall stopping distance can be worked out using the following equation:

overall stopping distance = thinking distance + braking distance

The table shows stopping distances from the Highway Code.

Speed (mph)	20	30	40	50	60	70
Thinking distance (m)	6	9	12	15		21
Braking distance (m)	6	14	24	38	56	75
Overall stopping distance (m)	12	23	36	53		96

- (a) Complete the table. [2]

- (b) (i) Describe how worn tyres affect the following distances. [2]

Thinking distance

Braking distance

- (ii) Describe how a driver using a mobile phone affects the following distances. [2]

Thinking distance

Braking distance

6



2. This question is about the Solar System.

(a) **Complete** the following sentence. [2]

The Solar System was formed by the collapse of a cloud of and
..... .

(b) The table shows some information about planets in our Solar System.

	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune
mean distance from Sun (AU)	0.4	0.7	1	1.5	5.2	9.5	19.2	30
mass compared with Earth	0.055	0.815	1	0.107	318	95	15	17
orbital time (years)	0.24	0.60	1	2	12	30	84	160
mean temperature at surface on sunny side (°C)	430	465	20	−20	−150	−170	−200	−210
number of moons	0	0	1	2	63	60	27	13

Use the information in the table to answer the questions that follow.

(i) Describe how the temperature on a planet depends on the distance from the Sun. [1]

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(ii) Identify the planet that does not follow this trend. [1]

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Examiner
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- (iii) Pallas is an asteroid found in the asteroid belt.
- I. Estimate its temperature °C [1]
- II. Estimate its orbital time years [1]
- (iv) It is suggested that the greater the mass of a planet, the more moons that will orbit around it. Explain whether the data agrees with this suggestion. [2]

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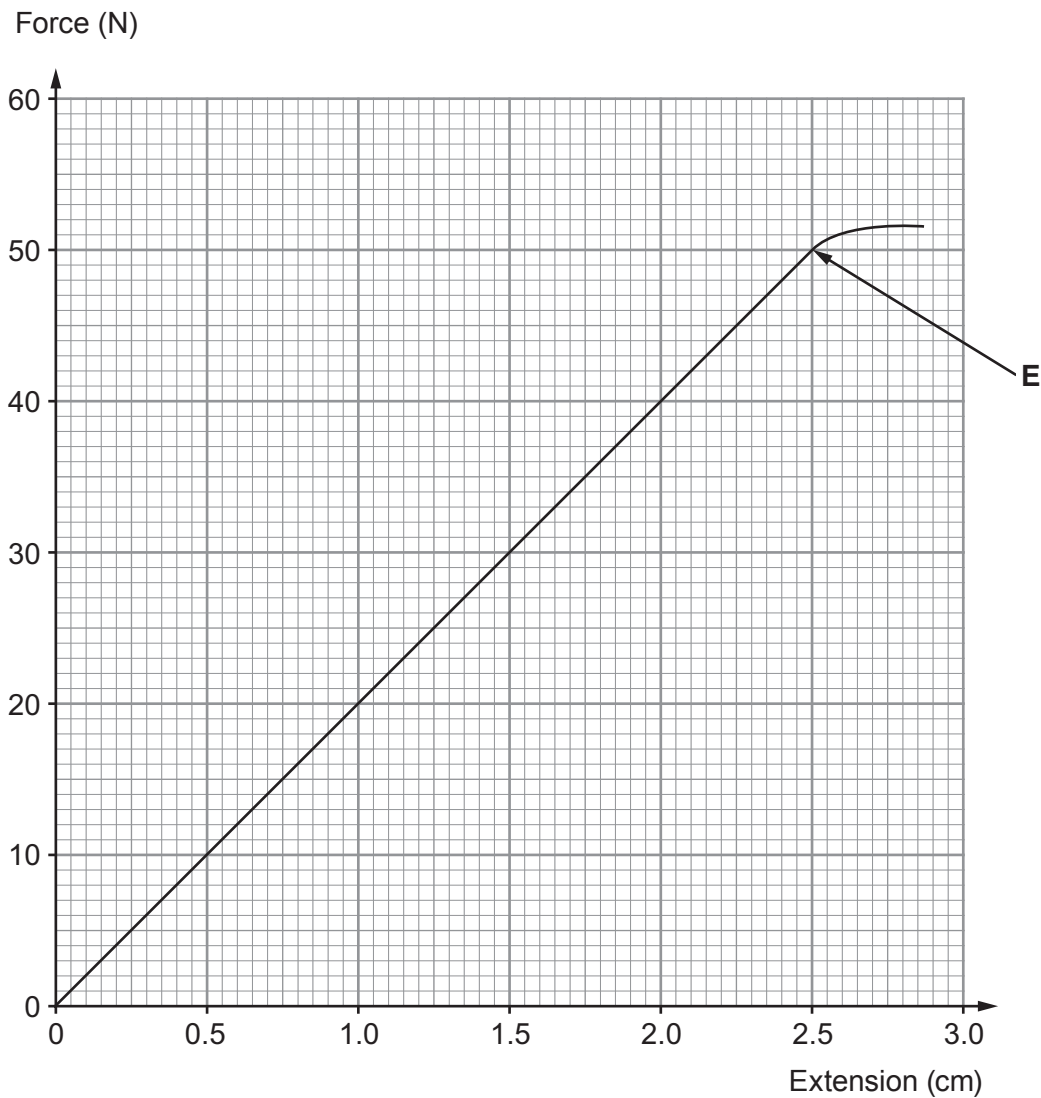


3. An engineering firm in South Wales makes springs for trampolines.



A trampoline is an elastic disc that is connected to a frame by many springs connected in parallel. As you land on a trampoline the springs stretch. The springs work to return to their normal length. The springs pull back against your weight as you land. The heavier the person on the trampoline, the longer the springs extend. Springs are tested to see if they return to their normal length after being stretched. The spring will be permanently stretched if it extends beyond the elastic limit (point **E**) where Hooke’s Law is no longer obeyed.

The graph shows how far a particular spring extends when forces are applied.



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(a) (i) Use the graph to find the largest force at which Hooke’s Law is still obeyed. [1]

Force = N

(ii) Each spring in one trampoline will experience a maximum force of 58 N.
Explain whether this particular spring is suitable for use in the trampoline. [2]

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(b) (i) Calculate the spring constant for this spring up to point E using the equation: [3]

spring constant = $\frac{\text{force}}{\text{extension}}$

Spring constant = N/cm

(ii) **Add a line** to the graph for a spring with a smaller spring constant. [2]

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4. One type of car which is very efficient is a hybrid electric vehicle, which has both a conventional fuel engine and an electric motor which runs from batteries.

Data about a hybrid electric / petrol car is given below.

Minimum time taken to accelerate from 0 - 30 m/s	12 s
Mass of car	1 100 kg
Mean CO ₂ emissions	90 g/km
Mean fuel economy	32 km/litre

(a) The car travels 160 km per week.

(i) Calculate the mean mass of CO₂ emitted by the car in a week. [1]

CO₂ = g

(ii) Calculate how many litres of fuel are used every week. [1]

Fuel used = litres

(b) (i) Use data from the table and an equation from page 2 to calculate the maximum acceleration of the car. [3]

Acceleration = m/s²



(ii) Use an equation from page 2 to calculate the size of the resultant force required to produce this acceleration. [2]

Resultant force = N

(c) State **one** other way cars are made more energy efficient. [1]

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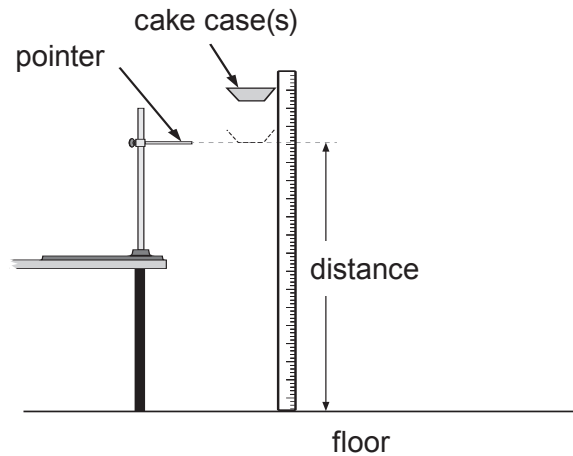
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- 5.** Students investigate the terminal speed of falling paper cake cases.



- (a) (i) State what is meant by terminal speed.

[1]

- (ii) Describe how you would carry out the investigation to find out how the mass of falling paper cake cases affects their terminal speed. [6 QER]

[6 QER]



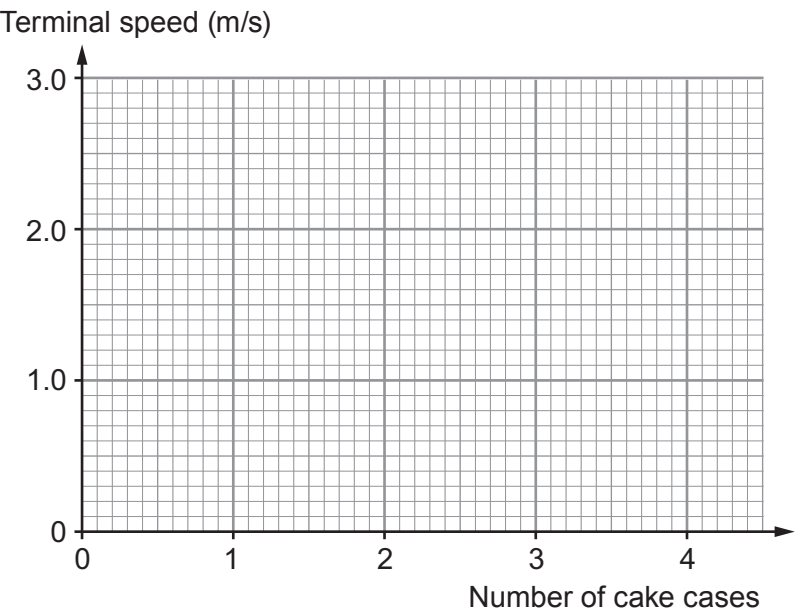
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(b) The table shows results from the investigation.

Number of cake cases	Distance (m)	Mean time taken for cake cases to fall 1.50 m (s)	Terminal speed (m/s)
0			0
1	1.50	0.90	1.7
2	1.50	0.68	2.2
3	1.50	0.60	
4	1.50	0.56	2.7

(i) **Complete the table** using an equation from page 2 to find the value of the missing terminal speed. *Space for workings.* [2]

(ii) Plot the data in the table on the grid below and draw a suitable line. [3]



(iii) One student suggests that doubling the number of cake cases reduces the **time** to fall by a half. Explain whether the results support this suggestion. [3]

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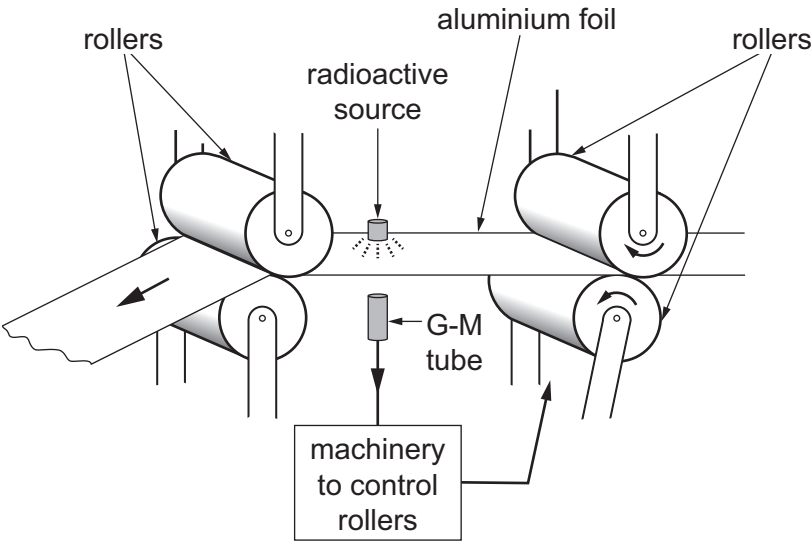
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6. A group of students study the uses of radioactivity. They find that radioactive isotopes are widely used in a variety of applications. For example, alpha emitters are used in smoke detectors. Medical diagnosis and cancer treatments use a range of radioisotopes emitting alpha, beta and gamma radiation.

One use they study in detail is monitoring the thickness of aluminium foil when it is manufactured. Radiation passes through the aluminium foil and is detected by a G-M tube. Changes to the thickness cause a difference in the count rate detected and adjustments can then be made to the pressure applied by the rollers.



Different radioisotopes have different half-lives and decay in different ways. The properties of some radioisotopes are given in the table below.

Isotope	Symbol	Half-life	Decay mode
strontium-90	$^{90}_{38}\text{Sr}$	29 years	beta
americium-241	$^{241}_{95}\text{Am}$	432 years	alpha
caesium-137	$^{137}_{55}\text{Cs}$	30 years	gamma
phosphorous-32	$^{32}_{15}\text{P}$	14 days	beta
actinium-225	$^{225}_{89}\text{Ac}$	10 days	alpha



Examiner
only

(a) Tick (✓) the boxes alongside the **three** correct statements below. [3]

- Alpha radiation consists of helium nuclei ☐
- Alpha radiation is more ionising than gamma ☐
- Strontium-90 has 90 nucleons ☐
- Gamma radiation consists of low energy waves ☐
- Gamma radiation only travels a short range in air ☐
- Beta radiation consists of slow moving electrons ☐

(b) One of the students suggests that strontium-90 is the most suitable isotope from the table for monitoring the thickness of aluminium foil. Explain whether or not you agree with this suggestion. [3]

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(c) (i) Explain what is meant by the statement: ‘The half-life of strontium-90 is 29 years.’ [2]

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(ii) Calculate the time taken for the activity of strontium-90 to fall to $\frac{1}{8}$ th of its initial value. [3]

Time = years



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(d) The students' teacher demonstrates experiments with radioactive sources. First she measures the radiation in the laboratory, recording 150 counts in 5 minutes. This allows her to work out the count rate of the background radiation.

(i) Calculate the background radiation count rate in counts per **second** (cps). [2]

Background count rate = cps

(ii) Suggest **two** ways in which the teacher could improve the accuracy of her result. [2]

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END OF PAPER

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SCIENCE (Double Award)



Unit 6 – PHYSICS 2
FOUNDATION TIER

WEDNESDAY, 22 MAY 2019 – AFTERNOON
1 hour 15 minutes

For Examiner's use only		
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2.	8	
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Equations

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acceleration = gradient of a velocity-time graph	
resultant force = mass $\times$ acceleration	$F = ma$
weight = mass $\times$ gravitational field strength	$W = mg$
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force = spring constant $\times$ extension	$F = kx$

SI multipliers

Prefix	Multiplier
m	1×10^{-3}
k	1×10^3
M	1×10^6



Answer all questions.

1. (a) The following diagrams show the life cycle of two different mass stars.
Diagram 1 shows how a star that is similar in mass to our Sun changes with time.
Diagram 2 shows the changes for a star that is 8 times more massive than our Sun.

Complete the labelling on diagrams **1** and **2**, using words or phrases from the box below.
Words or phrases can be used once, more than once or not at all. [4]

asteroid	neutron star	red giant	supergiant	white dwarf
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Diagram 1 – Star of mass similar to our Sun

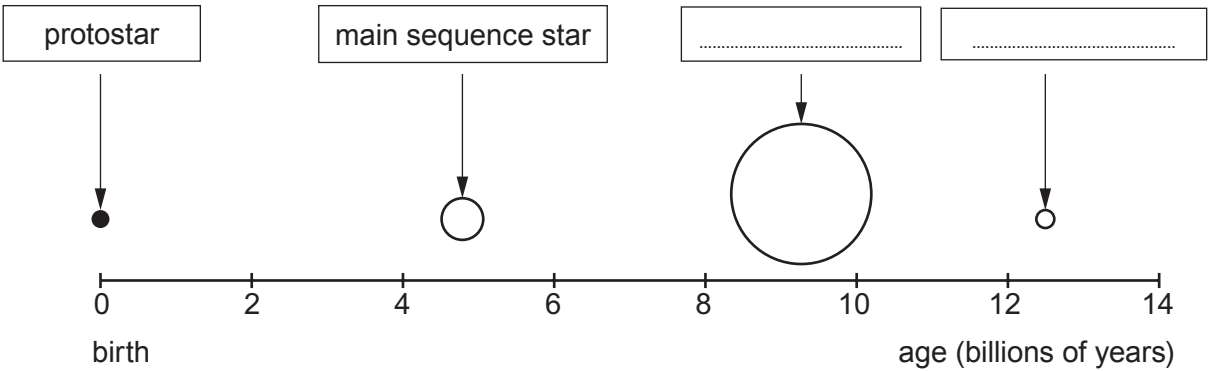
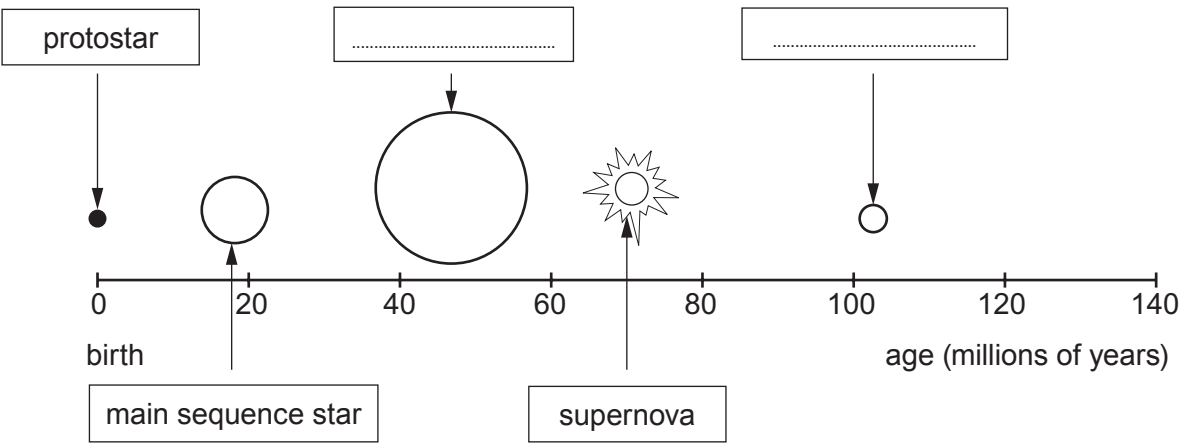


Diagram 2 – Star that is 8 times more massive than our Sun



(b) **Underline** the word or phrase in the brackets which correctly completes each sentence. [4]

- (i) A main sequence star is stable because its gravitational force is
(**less than** / **equal to** / **greater than**) the force caused by its radiation pressure.
- (ii) Stars generate their energy by the (**burning** / **fusion** / **fission**) of increasingly
heavier elements.
- (iii) During the final stages in the life cycle of some stars heavy elements are ejected
when the star becomes a (**gaseous giant planet** / **supernova** / **supergiant**).
The collapse of a cloud of gas and dust combined with these heavy elements can
eventually form a (**Solar System** / **galaxy** / **Universe**).

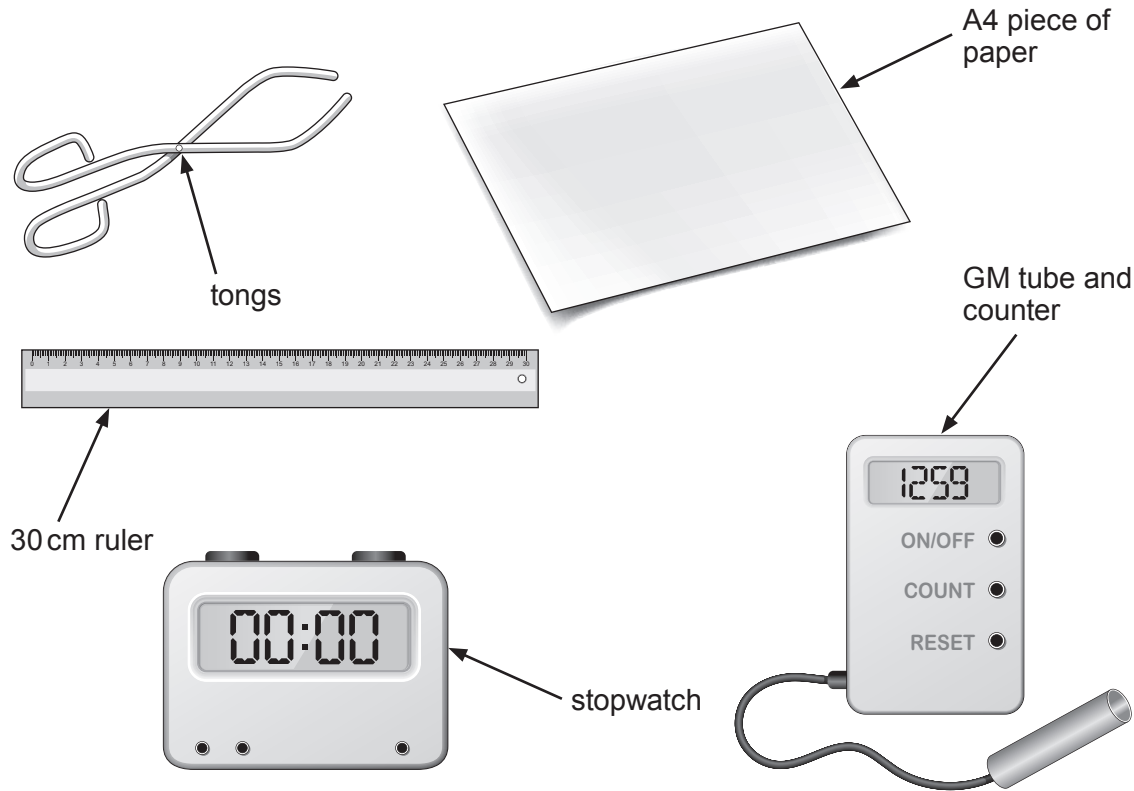


2. (a) Draw **one** line from each of the named radiations to the correct description. [2]

Named radiation	Description
alpha radiation	electromagnetic waves
beta radiation	helium nuclei
gamma radiation	high energy electrons

(b) A laboratory has 3 different radioactive sources that are not labelled. It is known that one of the sources emits **only alpha** radiation. Another of the sources emits **only beta** radiation. The third source emits **both beta and alpha** radiation.

Only the following apparatus is available.



Examiner
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Explain how the apparatus can be used to collect data **and** how the data can be analysed to identify the 3 sources. [6 QER]

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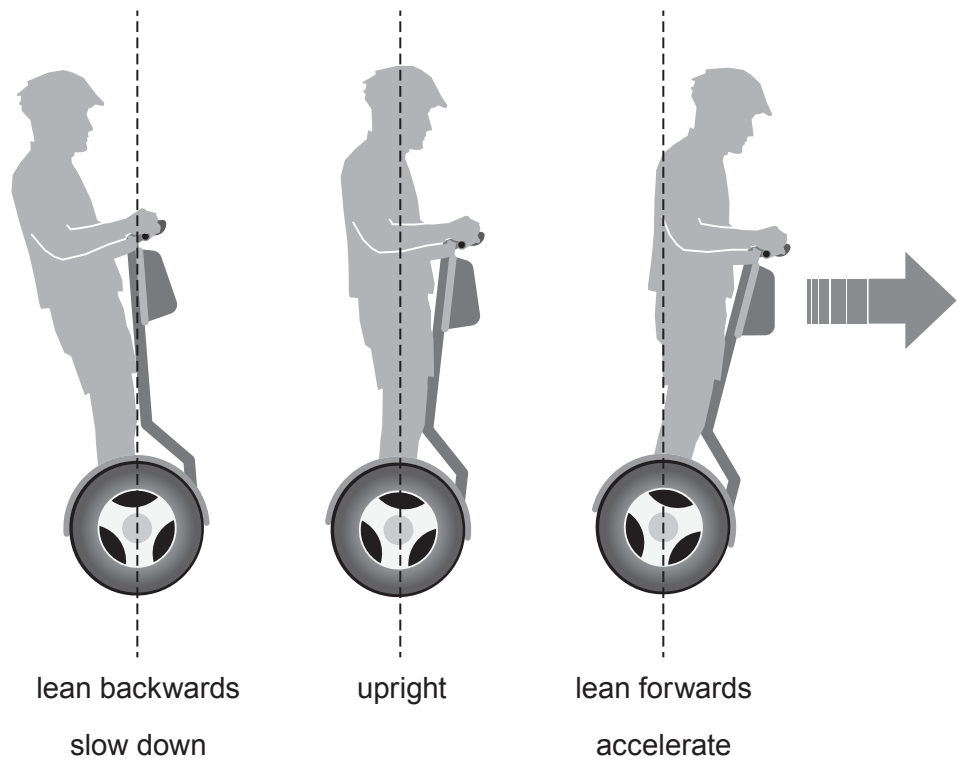
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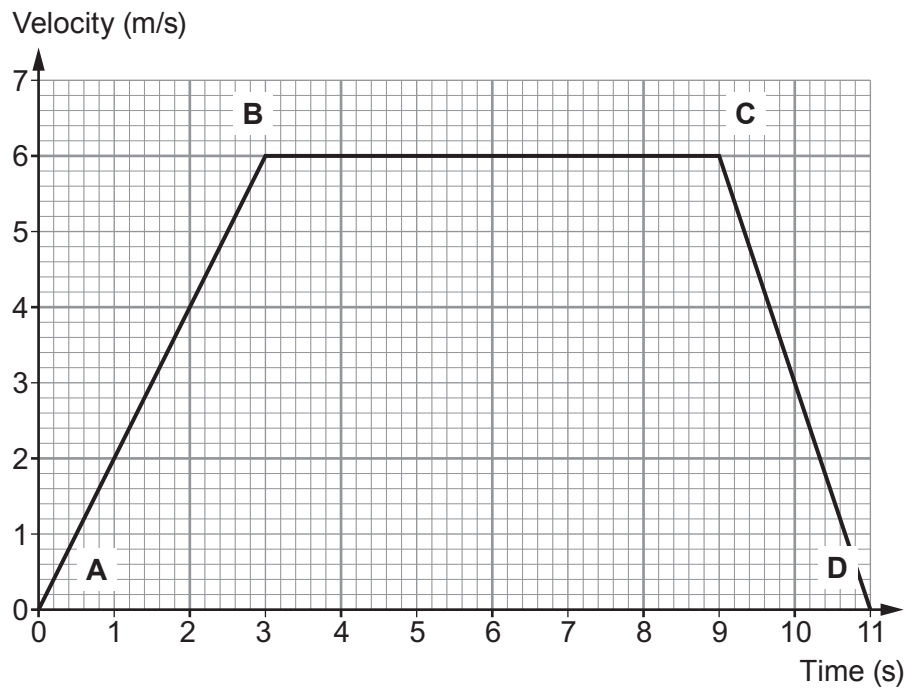
3430U601
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3. A Segway vehicle can be used as a method of transport for a person. The person leans forwards to make the Segway accelerate and leans backwards to slow it down. It is recommended that short training sessions are carried out before using the Segway on a busy pavement.



The velocity-time graph below relates to a short Segway training session that is carried out on a **dry** pavement. The person riding the Segway is instructed to carry out an emergency stop during the training.



Use information on the graph opposite to answer the following questions.

- (a) Between which points, **A to B**, **B to C** or **C to D**, is the person leaning forward on the Segway? [1]

- (b) Use the equation:

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

to calculate the acceleration during the first 3 seconds and state its unit. [3]

Acceleration =

Unit =

- (c) The combined mass of the person and the Segway is 110 kg.
Use the equation:

$$\text{resultant force} = \text{mass} \times \text{acceleration}$$

to calculate the resultant force during the first 3 seconds. [2]

Resultant force = N

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- (d) The reaction time of the person riding the Segway is 0.6 seconds.
Use the graph on page 8 to determine the time, during the training session, at which the rider is asked to carry out the emergency stop. [2]

Time = s

- (e) Newton’s first law of motion states “An object will remain at rest, or move at constant velocity, if the forces acting on it are balanced”.

A student states that during the Segway training session balanced forces act on it for 11.0 s. Explain whether you agree with the student. [2]

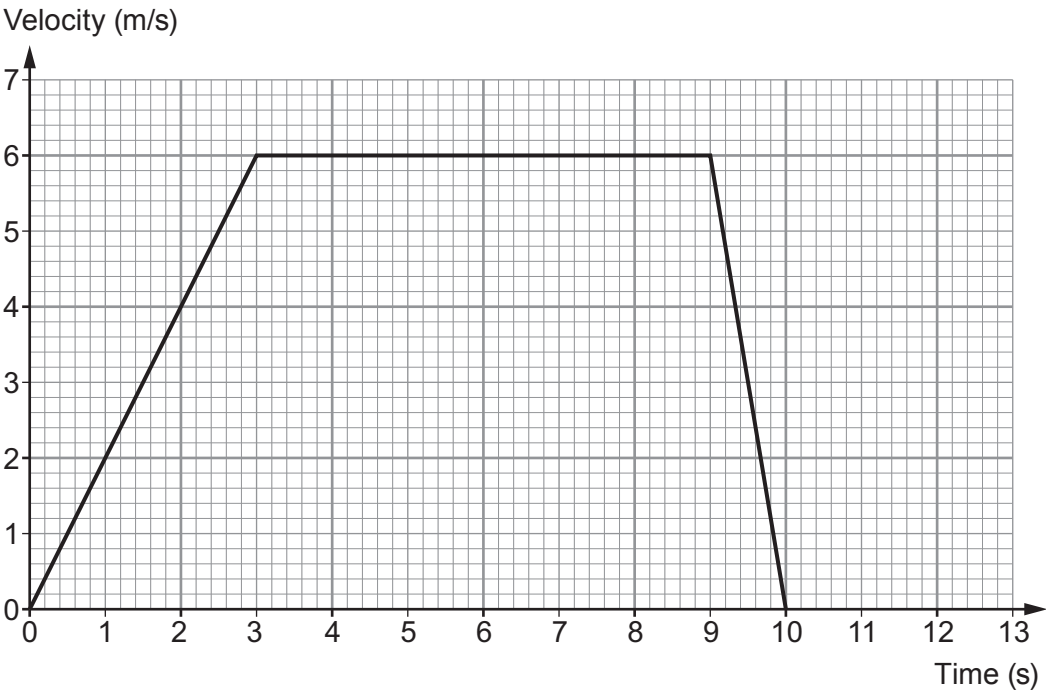
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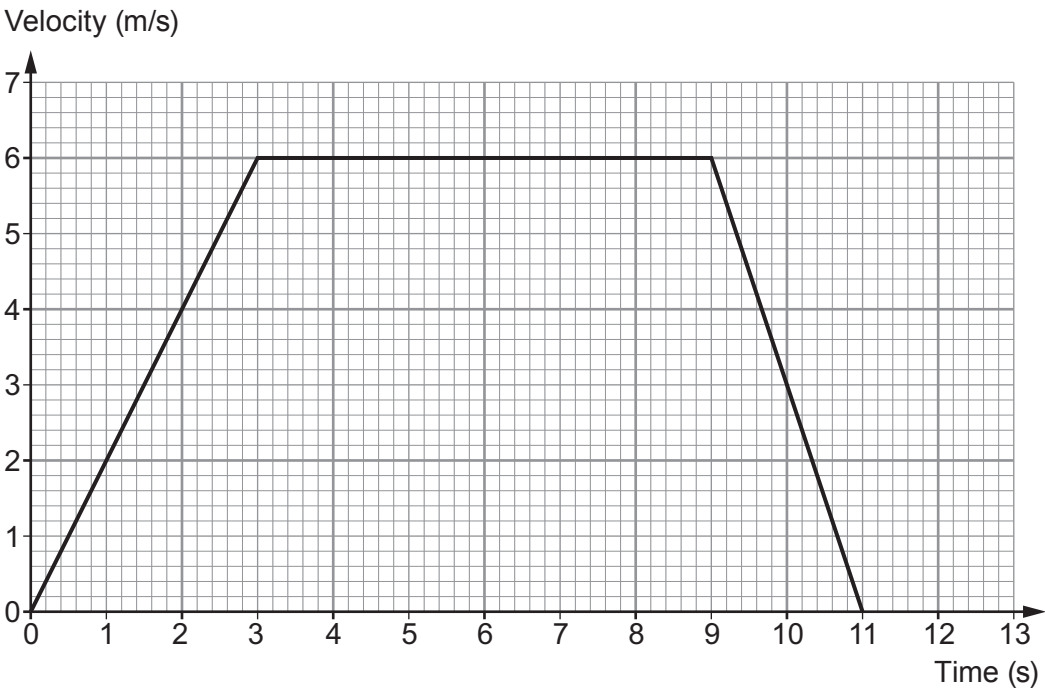
- (f) The graphs below and opposite show 3 possible training sessions carried out by the same Segway rider. Lucy says that **Graph C** would represent the emergency stop carried out on an **icy** pavement. Explain whether you agree with Lucy. [2]

Graph A

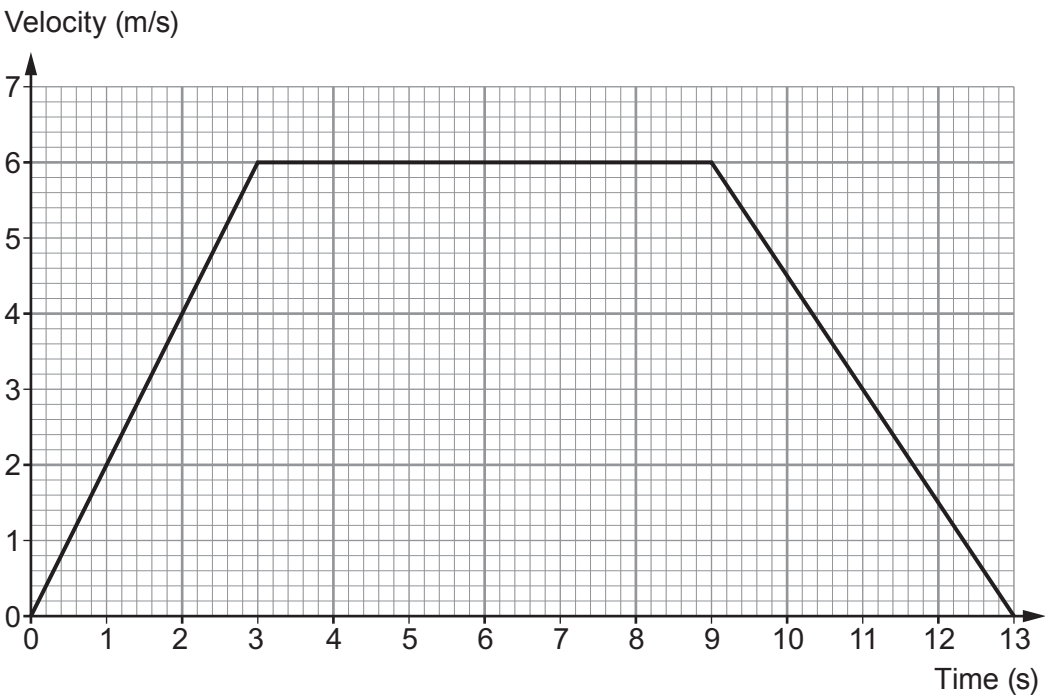


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Graph B



Graph C



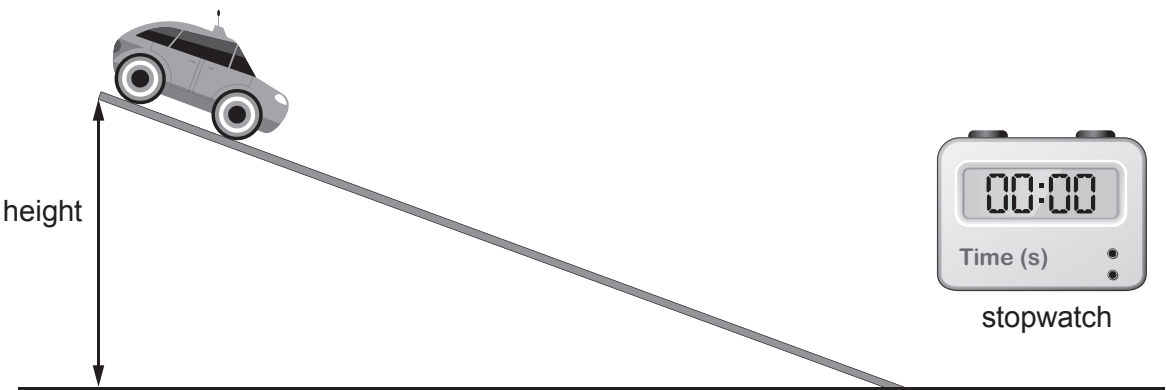
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4. Two students carry out an experiment with a toy car and a 2.50 m long piece of track.



They investigate how changing the height at one end of the track affects the time taken for the toy car to travel down 2.50 m of the track. One student releases the car and the other uses a stopwatch to measure the time. They do this 3 times for each height. Their results are shown in the table.

Height (cm)	Distance travelled (m)	Time (s)			
		Result 1	Result 2	Result 3	Mean
10	2.50	4.0	4.1	4.0	4.0
20	2.50	2.9	3.3	3.1	3.1
30	2.50	2.5	2.7	2.4	2.5
40	2.50	2.1	2.0	2.3	2.1
50	2.50	2.0	1.8	1.9	1.9

(a) Identify a controlled variable in the table.

[1]

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(b) (i) Use the equation:

mean speed = $\frac{\text{distance travelled}}{\text{mean time}}$

to calculate the mean speed of the toy car when the slope is set at a height of 10 cm. [2]

Mean speed = m/s

(ii) I. Describe how the mean time changes as the height increases by 10 cm steps. [2]

.....
.....

II. Describe how the mean speed changes as the height increases. [1]

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(c) (i) One student says, "The most repeatable data is for a height of 50 cm". Explain why this statement is **incorrect** and write a similar correct statement. [2]

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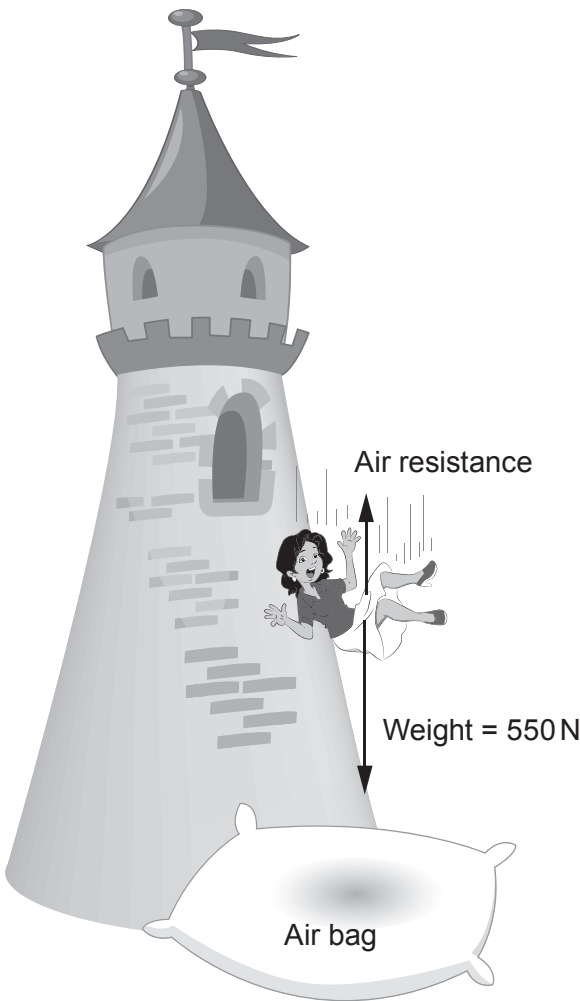
(ii) Explain why using a timer connected to light gates positioned at the start and end of the 2.50 m track will improve the results. [2]

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10



5. As part of a charity event a woman jumps from a tall tower and she safely lands on a large air bag that is beneath.



- (a) On Earth, an object of 1 kg has a weight of 10 N.
Calculate the mass of the woman.

[1]

Mass = kg



Examiner
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(b) The woman hits the air bag with 5.60 kJ of kinetic energy.
The airbag stops her in a distance of 2.8 m.

(i) Convert 5.60 kJ into joules. [1]

Kinetic energy = J

(ii) Use the equation:

$$\text{mean force} = \frac{\text{work done}}{\text{distance}}$$

to calculate the mean force bringing the woman to rest. [2]

Mean force = N

(iii) It is suggested that it would be safer if the air bag were filled with water. The distance to stop the same woman falling from the same tower would be 0.8 m instead of 2.8 m. Without further calculation explain whether you agree with the suggestion. [3]

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6. (a) Pupils in a class were given 200 coins to use in an experiment to simulate radioactive decay. They were asked to shake the coins in a bag, throw them out on the table then remove those showing “heads”. The number removed were counted and recorded in a table. The remainder of the coins were put back in the bag and the process was repeated again and again. Their results are shown in the table below.

Throw number	Total number of coins removed	Number of coins remaining
0	0	200
1	104	96
2	149	51
3		26
4		20
5		6
6		4

- (i) **Complete the table.** [1]
- (ii) After two throws, the number of coins remaining was 51. How many coins would you have **expected** to remain? [1]
.....
- (iii) After how many throws would the number of remaining coins fall to about **one eighth** of the original number? [1]
.....
- (b) Carbon-14 is a radioactive form of carbon that is present in all living material. Each nucleus of carbon-14 undergoes radioactive decay by emitting a beta particle to form nitrogen-14 according to the following decay equation, which is incomplete.

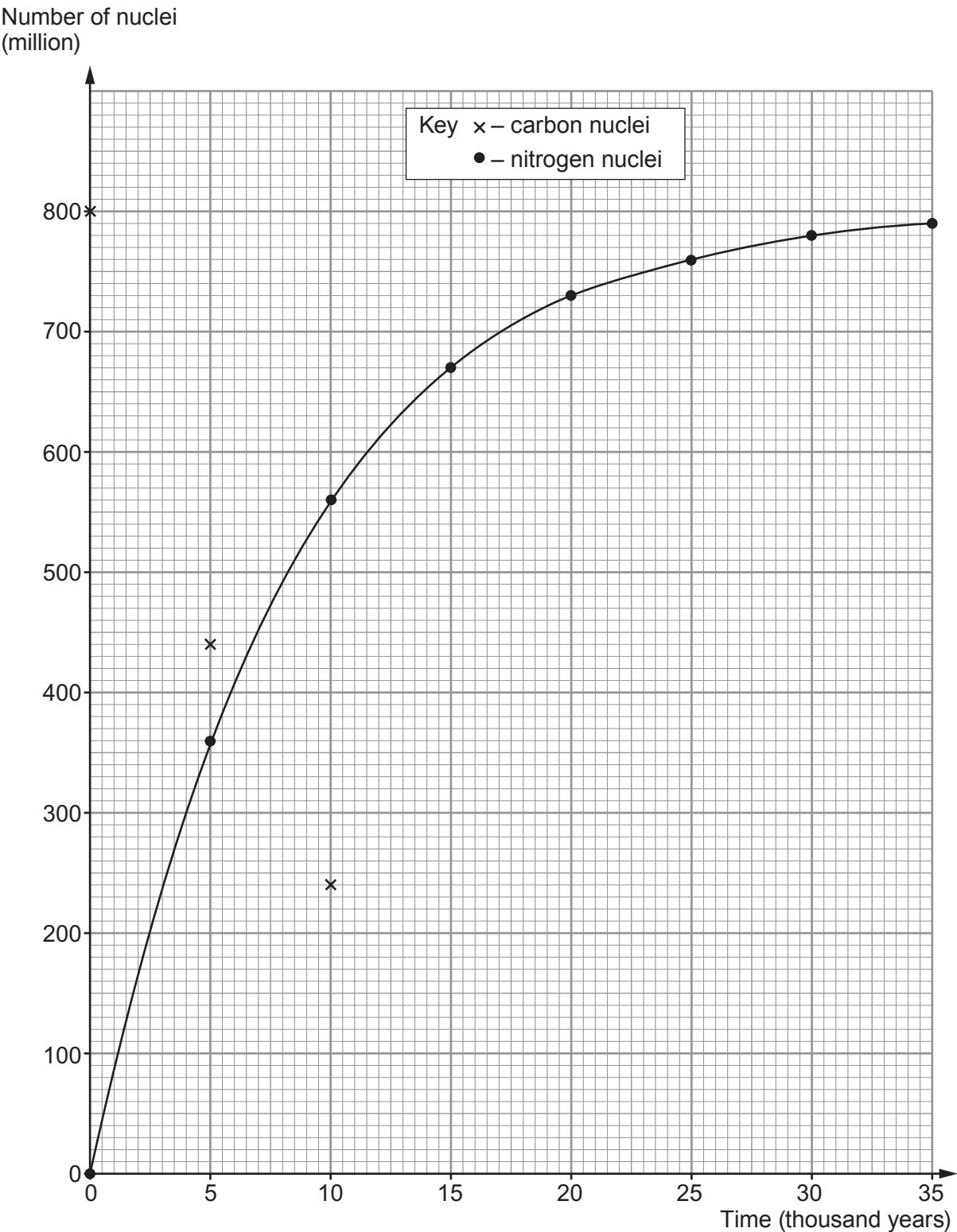


Complete the nuclear equation above.

[2]



- (c) A sample of 800 million carbon nuclei decays to create nuclei of nitrogen. The decrease in the sample creates an **increase** in the number of nuclei of nitrogen according to the following graph.



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(i) Complete the following table. [2]

Time (thousand years)	Total number of nuclei (million)	Number of nitrogen nuclei (million)	Number of carbon nuclei (million)
0	800	0	800
5		360	440
10		560	240
15		670	
20		730	
25		760	
30		780	
35		790	

(ii) On the grid opposite, plot points showing the **decay** of the carbon-14 nuclei. The first three crosses showing the numbers of carbon nuclei have been plotted for you.
Draw a suitable line. [3]

(d) (i) State the meaning of “the half-life” of a radioactive substance. [2]

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(ii) Use the graph to determine the half-life of carbon-14. [1]

Half-life = thousand years

THIS QUESTION CONTINUES ON PAGE 20



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- (e) Carbon dating is used to find the age of some ancient objects because carbon-14 is present in all once-living material. The process has been used to identify the age of the Turin shroud. This is a sheet of cloth that was claimed to be about 2000 years old. Three independent radiocarbon dating tests, carried out recently, attempted to identify the age of the cloth.



Out of **80 million** carbon-14 nuclei which were present in each sample of the original cloth, around 6 million have decayed into nitrogen. Use this information to explain whether the claim about its age is correct. [2]

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END OF PAPER

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Surname	Centre Number	Candidate Number
First name(s)		0



GCSE
3430U60-1



WEDNESDAY, 8 JUNE 2022 – AFTERNOON

SCIENCE (Double Award)

Unit 6 – PHYSICS 2
FOUNDATION TIER

1 hour 15 minutes

For Examiner's use only		
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JUN223430U60101

Equations

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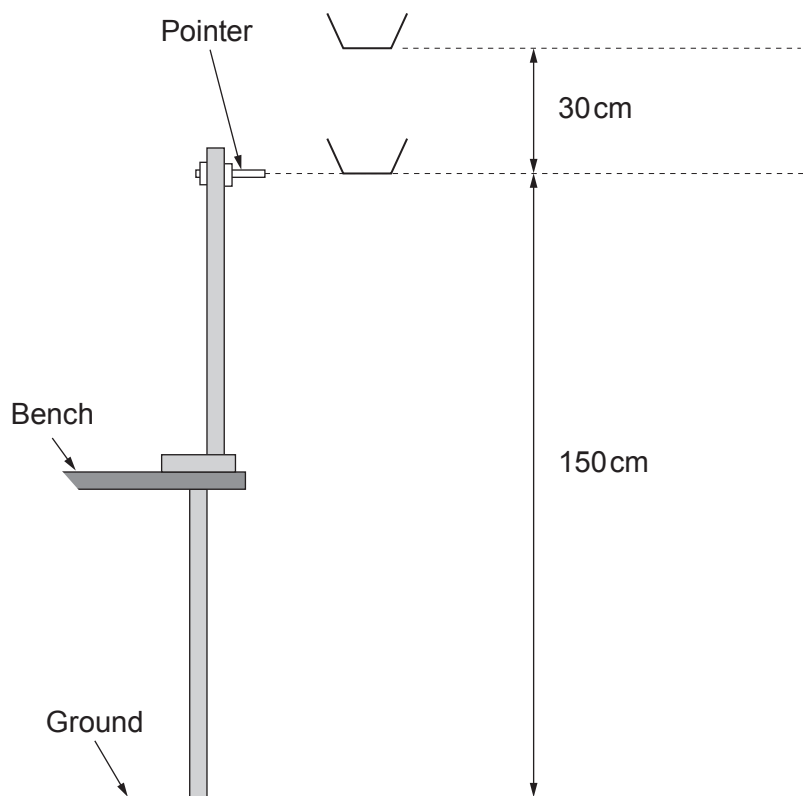
SI multipliers

Prefix	Multiplier
m	1×10^{-3}
k	1×10^3
M	1×10^6



Answer **all** questions.

1. A group of students investigate the terminal speed of a number of cake cases. They set up the following apparatus. They drop the cake cases from rest at a point 30 cm above a pointer. They measure the time taken for the different numbers of cake cases to fall 150 cm, from the pointer to the ground.



- (a) Complete the sentences below by underlining the correct phrase in the bracket. [3]
- (i) In this experiment, the independent variable is the
(**number of cake cases** / time of fall / height of pointer).
- (ii) In this experiment, the dependent variable is the
(**number of cake cases** / time of fall / height of pointer).
- (iii) During the first 30 cm of the fall, the cake cases
(**speed up** / slow down / fall at constant speed).



Examiner
only

(b) Five separate groups in the class carry out the same experiment. Their results and calculated values are shown below.

Distance (cm)	Number of cake cases	Time of fall (s)						Terminal speed (m/s)
		Group A	Group B	Group C	Group D	Group E	Mean	
	0							0
150	1	0.86	0.88	0.91	0.90	0.85	0.88	1.7
150	2	0.72	0.72	0.67	0.65	0.66	0.68	2.2
150	3	0.60	0.59	0.62	0.60	0.61	0.60	2.5
150	4	0.55	0.56	0.55	0.75	0.58		2.7
150	5	0.50	0.55	0.51	0.54	0.50	0.52	2.9

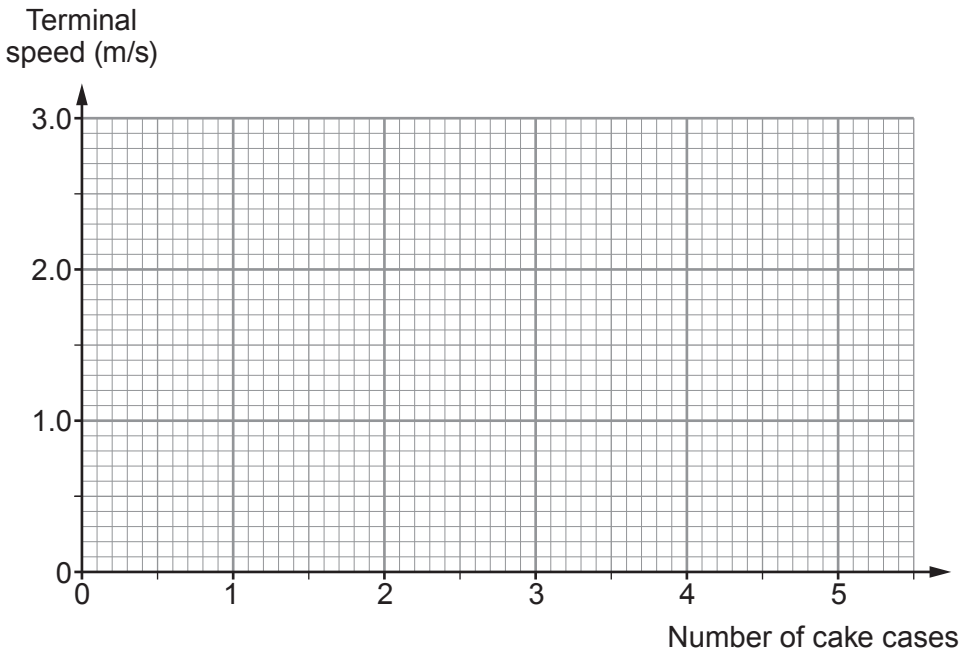
- (i) **Draw a circle** around the anomalous result when **4 cake cases** are dropped. [1]
- (ii) Calculate the mean time for 4 cake cases to fall 150 cm. [2]

Mean time = s



- (c) (i) Use the table below to plot the data on the grid and draw a suitable curve of best fit. [3]

Number of cake cases	Terminal speed (m/s)
0	0
1	1.7
2	2.2
3	2.5
4	2.7
5	2.9



- (ii) Complete the sentence below by underlining the correct phrase in each bracket. [2]

As the number of cake cases increases, the terminal speed (**increases / decreases / stays the same**) at (**an increasing / a decreasing / a constant**) rate.



2. The total stopping distance for a moving car is given by the equation below:

total stopping distance = thinking distance + braking distance

- (a) These distances may be affected by a number of factors.
Three of these factors are given in the table below.

Put a **tick** (✓) or a **cross** (X) in each box below to show whether the distance is affected by each factor. [3]

The first row has been done for you.

Factor	Thinking distance	Braking distance	Total stopping distance
Worn tyres	X	✓	✓
Drunk driver			
Wet road			

- (b) At a speed of 13 m/s, the thinking distance of an alert driver is 9.1 m.

Use the equation:

time = $\frac{\text{distance}}{\text{speed}}$

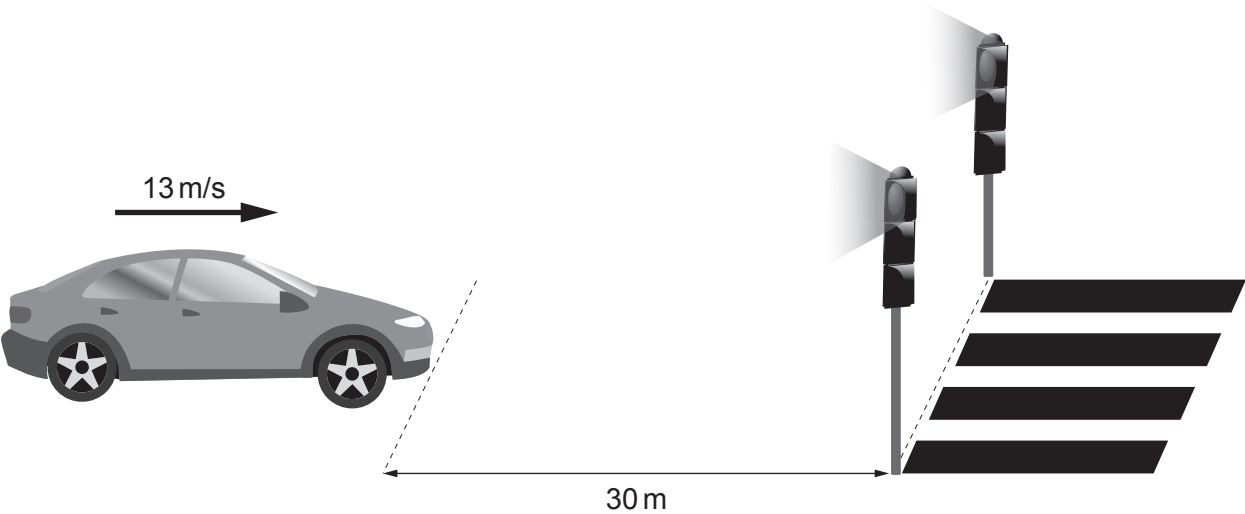
to calculate the thinking time.

[2]

Thinking time = s



- (c) A driver of a car travelling at 13 m/s sees traffic lights 30 m ahead when the lights turn to red.



The thinking distance = 9.1 m and the braking distance = 13.9 m at this speed.

Use the equation:

total stopping distance = thinking distance + braking distance

to explain whether the car would be able to stop before reaching the crossing. [2]

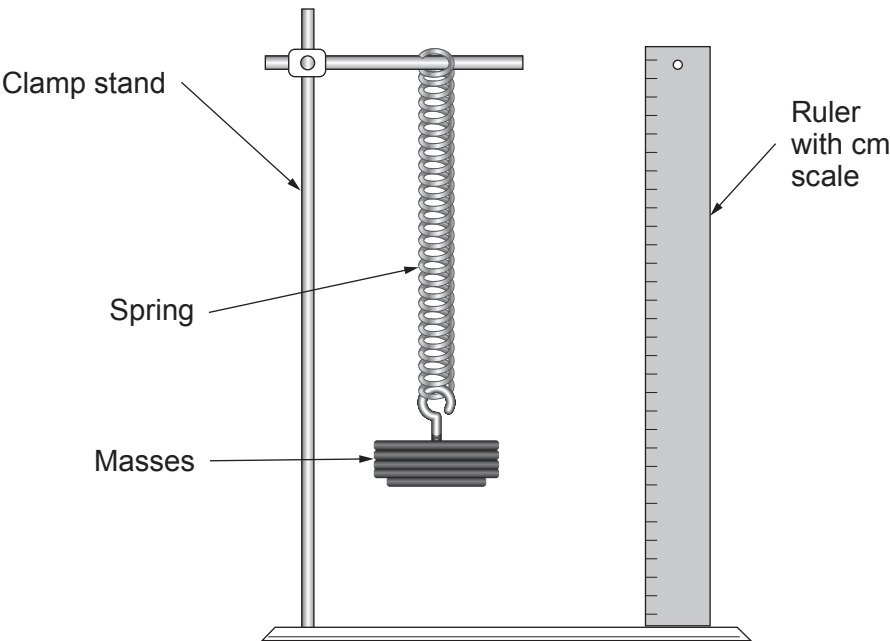
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7



3. The diagram shows a spring being stretched by adding masses. The spring is stretched within its elastic limit.



- (a) Part of the method used for investigating the extension of the spring is shown below.

METHOD

- 1. Record the original length of the spring.
- 2. Suspend the spring from the clamp and attach the 100 g mass hanger.
- 3.
- 4. Add a further 100 g to the spring and record the new length.
- 5.
- 6. Remove all masses and repeat steps 1–5 once more.
- 7.
- 8.

Complete the method by writing the letters **A, B, C, D** from the table below into the correct gaps in the method above.

[3]

A	Calculate the mean length for each mass added.
B	Record the new length of the spring.
C	Repeat step 4 until a total mass of 700g has been added.
D	Calculate the mean extension for each mass added.



Examiner
only

(b) One source of inaccuracy in this experiment is measuring the length of the spring. State **two** ways this measurement could be improved. [2]

- 1.
- 2.

(c) The spring has an extension of 8 cm when a mass of 400 g is suspended from it.

(i) State the weight of the 400 g mass. [100 g weighs 1 N] [1]

Weight = N

(ii) Use the equation:

spring constant = $\frac{\text{weight}}{\text{extension}}$

to calculate the spring constant for an extension of **8 cm**. [2]

Spring constant = N/cm

(iii) John says that a mass of 200 g would stretch the spring by 2 cm. Explain whether his claim is correct. [2]

.....

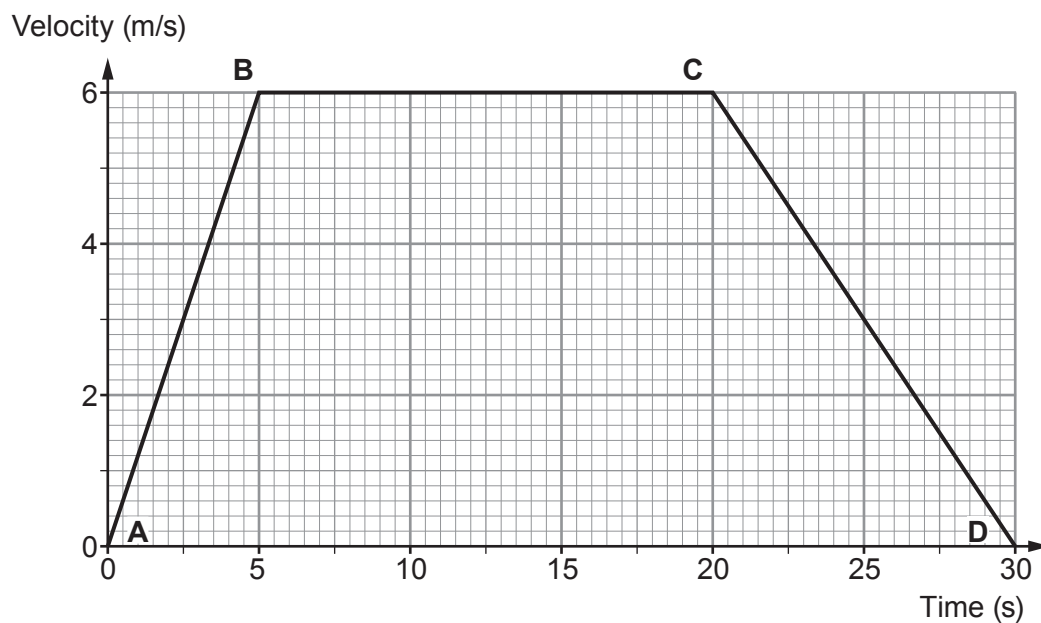
.....

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Examiner
only

4. The velocity-time graph shows the motion of a car for a time of 30 s.



Use data from the graph to describe the motion of the car.
No calculations are required.

[6 QER]

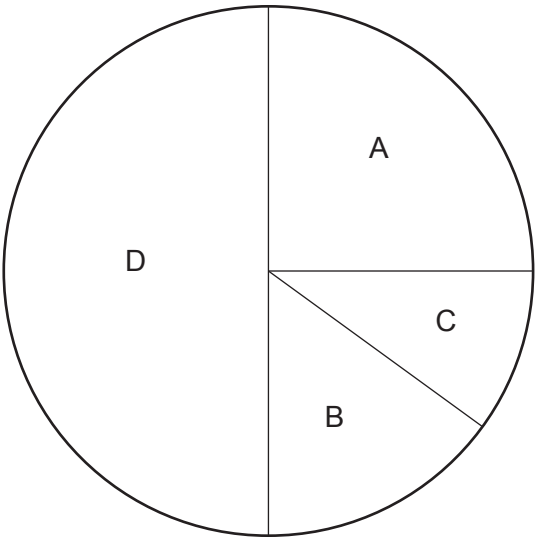
	6



5. (a) The pie chart shows the percentages of background radiation from various sources for an area of Cornwall in the UK.

(i) Use the following information to **complete the key** for the pie chart. [3]

Around 50 % of background radiation comes from radon gas.
25 % comes from ground and buildings and medical together.
15 % comes from cosmic rays.
10 % comes from food and drink.



Key for pie chart

Source of background radiation	Letter on pie chart
Radon gas	
Food and drink	
Ground and buildings and medical	
Cosmic rays	

(ii) The background radiation from radon gas in Mid Wales **is smaller** than in Cornwall.
The background radiation from all other sources remains the same.

Alan states that the pie chart for Mid Wales would be the same as the one above.
Explain whether you think that Alan is correct. [2]

.....

.....

.....



(b) The isotope polonium-210 has a half-life of 140 days.
It is a source of alpha radiation.
200 g of polonium-210 is stored in a locked cupboard.

(i) The mass of the polonium-210 halves in 140 days.
State **one** other quantity that halves in that time.

[1]

(ii) **Complete the table** below to show how the mass of polonium remaining changes with time.

[2]

Time (days)	Mass of polonium remaining (g)
0	200
140	100
280	
420	

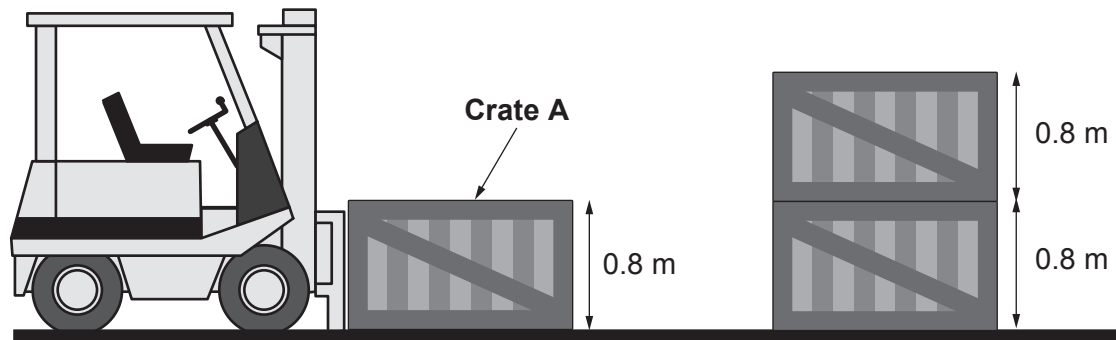
(iii) **Complete the following table**, giving the numbers of particles in a nucleus of polonium-210 ($^{210}_{84}\text{Po}$).

[3]

Name of particles	Number of particles in the nucleus
nucleons	
protons	
neutrons	



6. On a farm a fork-lift truck is used to stack wooden crates of Welsh apples.



- (a) (i) Each crate of apples has a weight of 450 N and is 0.8 m high.
Use an equation from page 2 to calculate the work done in putting **crate A** on top
of the **2 other** crates. [3]

Work done = J

- (ii) State the gain in potential energy (PE) of the crate when lifted on to the stack. [1]

Gain in PE = J

- (iii) Give a reason why the fork-lift truck uses more energy lifting crate A than the work
done calculated in part (i). [1]

.....
.....

- (b) Each crate of apples has a weight of 450 N.
The mass of the empty crate is 12 kg.
Calculate the mass of apples contained in the crate.
(On Earth, an object of weight 10 N has a mass of 1 kg.) [2]

Mass of apples = kg



7. Jupiter has 79 moons in orbit around it. The table shows data for 4 of its moons.

Name of moon	Mean diameter (km)	Mean temperature (°C)	Orbit radius (km)	Orbit time (days)
Io	3 660	−163	421 700	1.8
Europa	3 120	−171	671 000	3.6
Ganymede	5 260	−163	1 070 400	7.2
Callisto	4 820	−139	1 882 700	16.7

Use information from the table above to answer the following questions.

- (a)

(i)

State which moon has the highest mean temperature.

[1]
- (ii)

For the solar system, as the orbit radius increases around the Sun, the mean temperature of the planets generally decreases.
Explain why the orbit radius of the moons around Jupiter does not affect their temperatures in the same way.

[2]

.....

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- (iii)

Peter states that as Callisto has the longest orbit time it must be the largest moon. Determine whether Peter's claim is correct.

[1]

.....

.....
- (iv)

Peter correctly notices that Ganymede has an orbit time that is exactly double Europa's.
Peter suggests that as the orbit time **doubles**, the orbit radius also **doubles**.
Use **only** the data for Ganymede and Europa to determine whether Peter's claim is true.

[1]

Space for calculations.

.....

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Examiner
only

- (b) (i) Two of the statements listed below are correct. One correct statement has already been ticked.
Tick (✓) one more box to show the other correct statement. [1]

An Astronomical Unit (AU) is the mean distance that separates the Earth and the Sun.

☒

A light year is a measurement of time.

☐

A light minute is the distance travelled by light in 60 seconds.

☐

A light year is smaller than a light second.

☐

- (ii) Both Earth and Jupiter travel in elliptical paths around the Sun.
As they orbit the Sun the closest distance between Jupiter and Earth is 588 000 000 km. This is equivalent to 3.92 AU.

Calculate the distance, in km, that separates the **Earth and the Sun**. [2]
(1 AU = Earth to Sun distance)

Distance = km

8

END OF PAPER



Surname	Centre Number	Candidate Number
First name(s)		0



GCSE
3430U60-1



THURSDAY, 25 MAY 2023 – MORNING

SCIENCE (Double Award)
Unit 6 – PHYSICS 2

FOUNDATION TIER

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	5	
2.	11	
3.	10	
4.	11	
5.	8	
6.	5	
7.	10	
Total	60	

ADDITIONAL MATERIALS

In addition to this examination paper, you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question 4(a).



JUN233430U60101

Equations

speed = $\frac{\text{distance}}{\text{time}}$	
acceleration [or deceleration] = $\frac{\text{change in velocity}}{\text{time}}$	$a = \frac{\Delta v}{t}$
acceleration = gradient of a velocity-time graph	
resultant force = mass $\times$ acceleration	$F = ma$
weight = mass $\times$ gravitational field strength	$W = mg$
work = force $\times$ distance	$W = Fd$
force = spring constant $\times$ extension	$F = kx$

SI multipliers

Prefix	Symbol	Conversion factor	Multiplier
milli	m	divide by 1000	1×10^{-3}
centi	c	divide by 100	1×10^{-2}
kilo	k	multiply by 1000	1×10^3
mega	M	multiply by 1 000 000	1×10^6



Answer **all** questions.

1. Our Sun is a low mass star. It is currently in the main sequence stage of its life.

(a) Use words from the box to complete the life cycle of our Sun. [3]

asteroid	protostar	supernova	white dwarf	red giant
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..... → main sequence → →

(b) Tick (✓) the **two** boxes next to the forces acting on our Sun that keep it stable. [2]

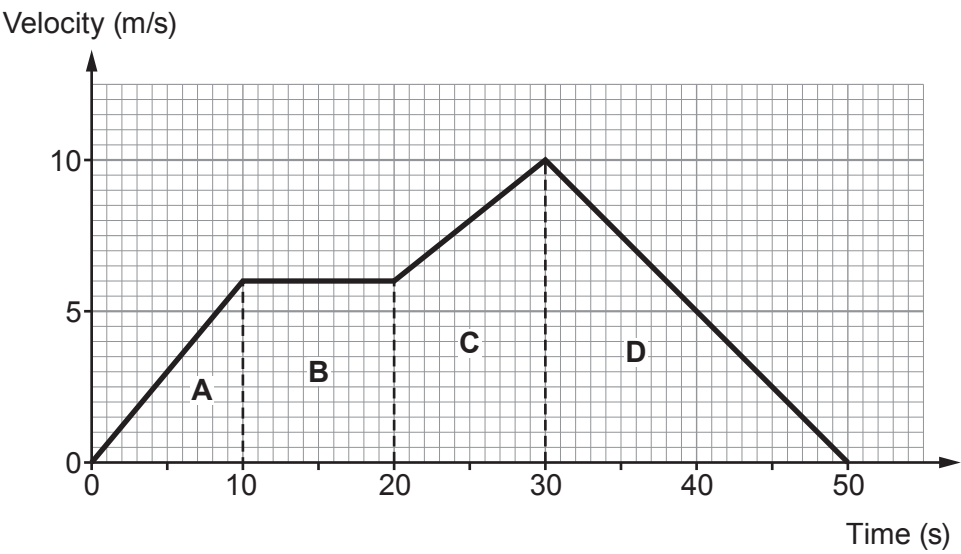
gas pressure	☐
friction	☐
magnetic	☐
gravity	☐

3430U601
03

5



2. The velocity-time graph is for part of a bus journey.



Use the information in the graph to answer the following questions.

- (a) **Complete the table** by placing **one** tick (✓) in each row to describe the motion in each region of the graph. Region A has been completed as an example. [3]

Region of graph	Not moving	Constant velocity	Accelerating	Decelerating
A			✓	
B				
C				
D				



(b) **Complete the following sentences** using numbers from the box. [3]

2	4	6	8	10	20	50
---	---	---	---	----	----	----

- (i) The maximum velocity of the bus is m/s.
- (ii) The change in velocity of the bus in region C is m/s.
- (iii) The bus accelerates for a **total time** of s.

(c) Use the equation:

acceleration = $\frac{\text{change in velocity}}{\text{time}}$

to calculate the acceleration in **region A**. [3]

acceleration = m/s²

(d) The bus travelled 270 m in the 50 s shown.

Use the equation:

mean speed = $\frac{\text{distance}}{\text{time}}$

to calculate the mean speed of the bus. [2]

mean speed = m/s

11



3. The picture shows a smoke detector on the ceiling of a room.
It contains the radioactive isotope americium-241 (Am-241) which emits alpha particles.



Smoke getting into the detector absorbs alpha particles.
This changes the electric current inside the detector and the alarm sounds.

- (a) Underline the phrase that correctly completes the sentence.

An alpha particle is a (**helium nucleus / hydrogen atom / fast moving electron**). [1]

- (b) Explain why the americium-241 inside the detector is not a risk to humans when the detector is used on the ceiling. [2]

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- (c) Explain why the smoke detector would not work if the radioactive isotope only emitted gamma rays. [2]

.....

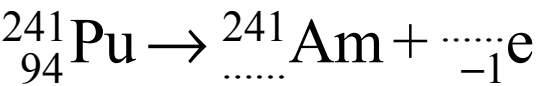
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- (d) Am-241 is produced when plutonium-241 (Pu-241) decays.
- (i) Select values from the box to answer the questions that follow.

94	95	147	0	241	2
----	----	-----	---	-----	---

- I. **Complete** the decay equation below. [2]



- II. State the number of **protons** in the nucleus of a Pu-241 atom. [1]

.....

- III. Calculate the number of **neutrons** in the nucleus of a Pu-241 atom. [1]

.....

- (ii) Circle the correct name for the other particle that is produced when Pu-241 decays. [1]

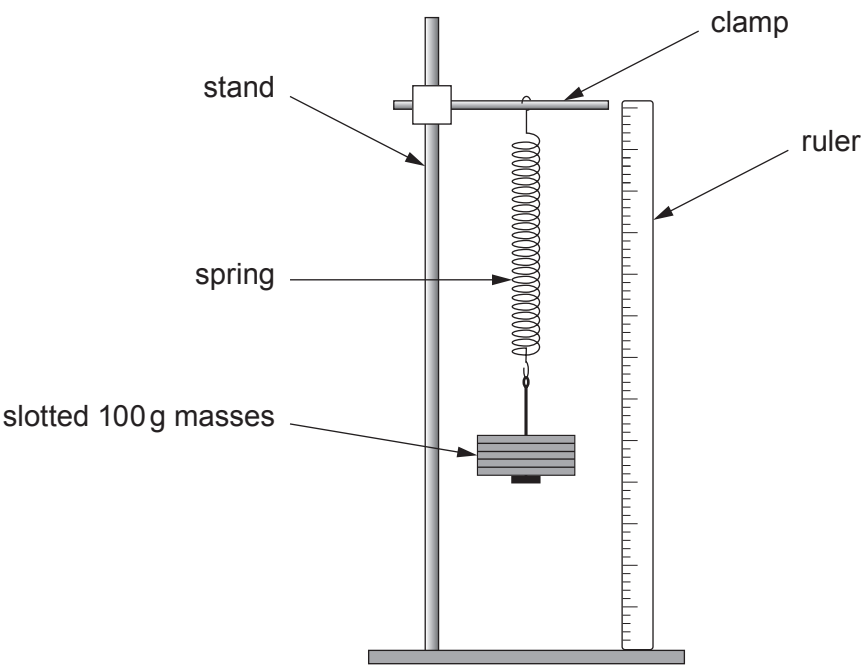
(gamma particle / beta particle / helium particle)

10

3430U601
07



4. (a) Describe how the apparatus shown below is used to investigate the extension of a spring. [6 QER]



- Include the following in your answer:
- how the apparatus is set up,
 - a description of the method used to obtain the results,
 - how the results are analysed.

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Examiner
only

(b) A spring is 3 cm long. When an object is added to it, its length increases to 18 cm.

(i) Calculate the extension produced by the object. [1]

extension = cm

(ii) The spring constant of the spring is 0.8 N/cm.
Use an equation from page 2 to calculate the force produced by the object on the spring. [2]

force = N

(iii) This force is the weight of the object.
Use the equation:

$$\text{mass} = \frac{\text{weight}}{\text{gravitational field strength}}$$

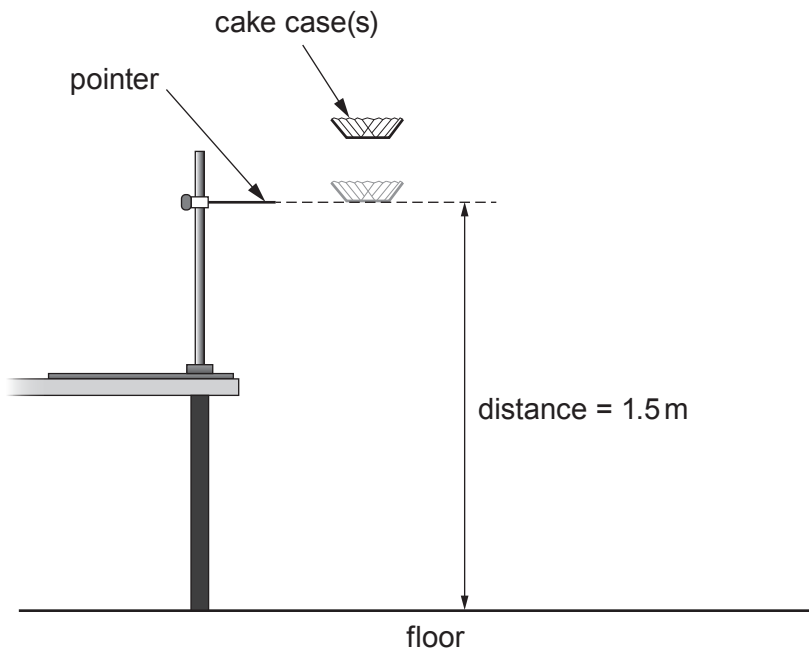
to calculate the mass of the object. (Gravitational field strength, $g = 10 \text{ N/kg}$) [2]

mass = kg

11



5. Students investigate the terminal speed of falling paper cake cases. The apparatus they use is shown below.



Their results are given in the table.

Number of cake cases	Time to fall (s)				Terminal speed (m/s)
	Trial 1	Trial 2	Trial 3	Mean	
1	0.85	0.94	0.91	0.90	1.7
2	0.68	0.62	0.65	0.65	2.3
3	0.58	0.62	0.57	0.59	2.5
4	0.52	0.26	0.54	0.44	3.4
6	0.44	0.48	0.46	0.46	3.3



Examiner
only

Use the results in the table to answer the following questions.

- (a)

(i)

Circle the anomalous result, in the table, for the time to fall.

[1]
- (ii)

Tom says that the mean time for 4 cake cases to fall is wrong.
Explain whether Tom is correct.
Space for calculation.

[2]

.....

.....

- (b)

Tom thinks that as the number of cake cases doubles the terminal speed doubles.
Explain whether you agree with Tom.

[2]

.....

.....

.....

- (c)

(i)

For 6 cases the times taken to fall are 0.44 s, 0.48 s and 0.46 s.

Use the equation:

uncertainty =

maximum value – minimum value

2

to calculate the uncertainty in the measurements of time for 6 cake cases to fall.

[2]

uncertainty = s

- (ii)

State **one** random error that is a cause of uncertainty when doing this experiment.

[1]

.....

8



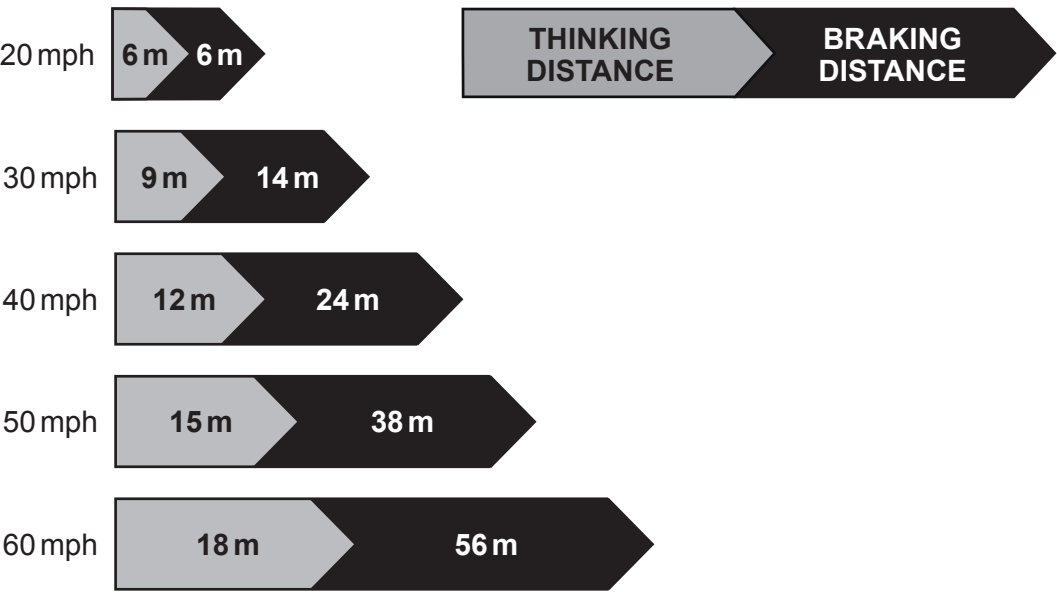
Examiner
only

6. (a) When a car stops the overall stopping distance is made up of two distances: the thinking distance and the braking distance.
Increasing speed increases both the thinking distance and the braking distance.

(i) State **one** factor, other than speed, which increases the thinking distance. [1]

(ii) State **one** factor, other than speed, which increases the braking distance. [1]

(b) The diagram gives information about stopping distances at different speeds.



On a dangerous road, it is proposed to reduce the speed limit from 40 mph to 20 mph.

Bethan makes the following 3 suggestions.

- 1. The thinking distance will halve.
- 2. The braking distance will halve.
- 3. The overall stopping distance will halve.

Explain whether you agree with each suggestion.
Include data from the diagram to support your answer. [3]

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5



7. (a) A class investigates radioactive decay. They model decay using **8-sided** dice.



Each of the 10 groups has 50 dice.
They throw the dice and remove all which land with an 8 facing upwards.
These represent decayed nuclei.
They repeat 7 more times and record the number of dice remaining after each throw.
Each group's results are then added together.

- (i) Freya suggests that it is good practice to add the results together to give a larger sample size.
Explain whether you agree. [1]

- (ii) The teacher calculates that after the first throw **around** 440 of the 8-sided dice should remain out of the 500.
Explain how she determined this number. [2]

- (b) The results from the experiment are given in the table below.

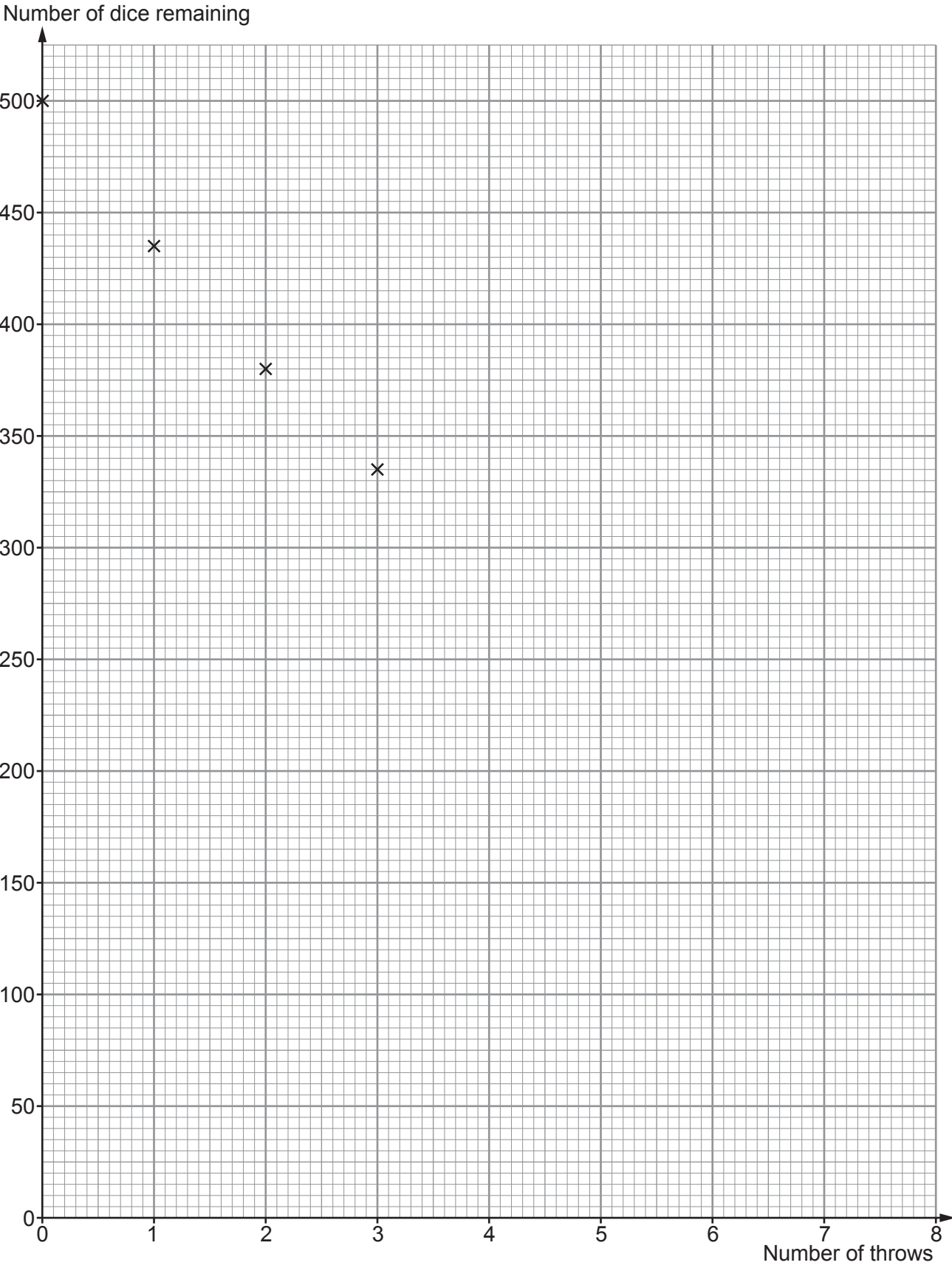
Number of throws	Number of dice remaining
0	500
1	435
2	380
3	335
4	291
5	256
6	224
7	196
8	171



(i) Plot the data on the grid below and draw a suitable curve.
The first 4 points have been plotted for you.

[3]

Examiner
only



Examiner
only

- (ii) **Add lines to the graph** to find the number of throws required to halve the number of dice. This is the half-life.
Give your answer to 1 decimal place. [2]

number of throws =

- (iii) The experiment is repeated with 10-sided dice.
leuan suggests the data could be used to model nuclear decay with a shorter half-life than with 8-sided dice.
Explain whether you agree. [2]

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10

END OF PAPER

